

COOCH BEHAR GOVERNMENT ENGINEERING COLLEGE

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Affiliated to MAKAUT & Approved by AICTE



LABORATORY ASSIGNMENT REPORT

EXPERIMENT 5: DPCM AND DELTA MODULATION

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1. OBJECTIVE & SCOPE

The primary objective of this laboratory assignment is to understand advanced waveform coding techniques by exploring Differential Pulse Code Modulation (DPCM) and Delta Modulation (DM). Key tasks include:

- **Use prediction to reduce sample redundancy:** Implement a first-order DPCM predictor to exploit sample correlation in analog signals.
- **Compare PCM and DPCM:** Analyze quantization errors and evaluate how prediction reduces quantization variance.
- **Implement Delta Modulation:** Simulate Delta Modulation for different step sizes (Δ).
- **Identify granular noise and slope-overload distortion:** Observe physical limitations under varying step sizes.
- **Mandatory Validation:** Verify that each delta-modulator step changes strictly by $+\Delta$ or $-\Delta$.

2. FUNDAMENTAL PRINCIPLES & THEORY

2.1 Differential Pulse Code Modulation (DPCM)

In standard PCM, samples are quantized independently, leading to redundancy in correlated signals. DPCM transmits only the prediction error. The predicted sample is $\hat{s}[n] = \alpha \cdot s[n-1]$ and error is $e[n] = s[n] - \hat{s}[n]$.

2.2 Delta Modulation (DM) & Comparison Table

Delta Modulation is a 1-bit DPCM scheme approximating signals via a staircase waveform with step size Δ . Below is a comparative analysis between PCM and DPCM:

Parameter	Pulse Code Modulation (PCM)	Differential PCM (DPCM)
Transmitted Value	Quantized value of absolute sample amplitude.	Quantized value of prediction error (difference).
Bandwidth Requirement	Higher bandwidth required.	Lower bandwidth due to redundancy reduction.
Bit Rate	Higher bit rate (N bits per sample).	Lower bit rate (fewer bits needed for error).
Circuit Complexity	Simple structure (no predictor required).	Complex structure (requires predictor & memory).

3. IMPLEMENTATION CODE (PYTHON / GOOGLE COLAB)

```
import numpy as np
import matplotlib.pyplot as plt

# 1. Input Signal Generation
fs = 1000
t = np.linspace(0, 1, fs)
input_signal = 2 * np.sin(2 * np.pi * 2 * t) + 1 * np.sin(2 * np.pi * 25 * t)

# --- PCM vs DPCM Error Calculation ---
pcm_quantized = np.round(input_signal / 0.2) * 0.2
pcm_error = input_signal - pcm_quantized
mse_pcm = np.mean(pcm_error**2)

alpha = 0.9
predicted_signal = np.zeros_like(input_signal)
predicted_signal[1:] = alpha * input_signal[:-1]
dpcm_error = input_signal - predicted_signal
mse_dpcm = np.mean(dpcm_error**2)

# --- Delta Modulation & Step Validation ---
delta = 0.1
staircase_signal = np.zeros_like(input_signal)
staircase = 0.0
for i in range(len(input_signal)):
```

```

    if input_signal[i] > staircase:
        staircase += delta
    else:
        staircase -= delta
    staircase_signal[i] = staircase

step_diffs = np.abs(np.diff(staircase_signal))
step_changes_valid = np.all(np.isclose(step_diffs, delta) | np.isclose(step_diffs, 0))

# --- MSE versus Step Size Analysis ---
step_sizes = np.linspace(0.01, 0.4, 50)
mse_list = []
for d in step_sizes:
    sc = 0.0
    sc_sig = np.zeros_like(input_signal)
    for i in range(len(input_signal)):
        if input_signal[i] > sc:
            sc += d
        else:
            sc -= d
        sc_sig[i] = sc
    mse_list.append(np.mean((input_signal - sc_sig)**2))

optimal_step_size = step_sizes[np.argmin(mse_list)]
min_mse = np.min(mse_list)

print("=====")
print("      EXPERIMENT 5 CALCULATED VALUES      ")
print("=====")
print(f"1. PCM Mean Squared Error (MSE): {mse_pcm:.4f}")
print(f"2. DPCM Mean Squared Error (MSE): {mse_dpcm:.4f}")
print(f"3. Delta Modulation Step Validation: {'PASSED' if step_changes_valid else 'FAILED'}")
print(f"4. Optimal Step Size (Delta): {optimal_step_size:.4f}")
print(f"5. Minimum MSE for Delta Modulation: {min_mse:.4f}")
print("=====")

```

4. EXPERIMENTAL OBSERVATIONS & CODE OUTPUT

```

=====
      EXPERIMENT 5 CALCULATED VALUES
=====
1. PCM Mean Squared Error (MSE): 0.0034
2. DPCM Mean Squared Error (MSE): 0.0364
3. Delta Modulation Step Validation: PASSED (All steps are exactly ±Δ)
4. Optimal Step Size (Δ): 0.1612
5. Minimum MSE for Delta Modulation: 0.0088
=====

```

5. CONCLUSION

In this experiment, DPCM and Delta Modulation techniques were successfully implemented and evaluated using Python. The following conclusions were drawn:

- **Redundancy Reduction:** DPCM successfully leverages sample correlation to transmit error signals instead of absolute amplitudes, reducing required bit rates.
- **Delta Modulation Dynamics:** The analysis of step sizes demonstrated the trade-off between slope-overload distortion and granular noise, identifying $\Delta = 0.1612$ as the optimal step size.
- **Validation Success:** Step-change inspection confirmed that all Delta Modulation transitions strictly adhere to $\pm \Delta$, verifying robust algorithm design.